

DERIVED EQUIVALENT GENTLE ALGEBRAS WHICH ARE NOT TILTING-COTILTING EQUIVALENT

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ABSTRACT. We give examples of two derived equivalent gentle algebras which are not tilting-cotilting equivalent.

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1. INTRODUCTION

Derived equivalence is one of the fundamental equivalence relations in the representation theory of finite-dimensional algebras. According to [8], an abundant supply of derived equivalences is provided by tilting-cotilting equivalences. It is therefore natural to ask: to what extent can derived equivalence be realized by tilting-cotilting equivalence? We also care about for which class of algebras these two notions of equivalences coincide.

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It is advisable to investigate this problem for gentle algebras, a class of algebras whose representation theory and derived classification are well-studied [2]. As shown in [4], for gentle algebras whose Gabriel quivers contain at most two cycles, derived equivalence turns out to be the same as tilting-cotilting equivalence. One might therefore suspect this for all gentle algebras. This paper proves that this expectation is false by constructing an explicit family of examples and analyzing their derived and tilting-cotilting behaviour.

Our main results are twofold. First, we introduce a family of gentle algebras $\Lambda(L, R)$ defined by two total orderings of a finite set $E = \{1, 2, \dots, n\}$, and show that for the special case $L = R = (1, 2, \dots, n)$, the algebra $\Lambda = \Lambda(L, L)$ is not affected by tilting-cotilting operations:

Theorem 1.1. *The algebra Λ is tilting-cotilting unique for $n \geq 1$.*

The proof proceeds in three steps. On one hand, we compute the Cartan matrix of Λ and show that it is congruent to I_n modulo 2. Second, using mutation invariance and string combinatorics, we show that the dimension vectors of the indecomposable summands of a tilting Λ -module T are congruent to the standard basis of $\mathbb{Z}^n$ modulo 2. the dimension vectors of the summands are congruent to the standard basis vectors modulo 2. Finally, we pass to trivial extensions [11] and use the derived equivalence classification of Brauer graph algebras [10] to show that every algebra obtained from Λ by a tilting or cotilting module must again be of the form $\Lambda(U, V)$. The theorem is proved by combining all these clues together.

On the other hand, we will show that Λ is far from being derived unique:

Theorem 1.2. *For every $n \geq 5$, there is an algebra Λ' derived equivalent to Λ but not tilting-cotilting equivalent to it.*

The proof uses the geometric model and complete derived invariants introduced by Amiot, Plamondon and Schroll in [2]. After giving the construction of Λ' , we compute their numerical invariants of [2, Theorem 7.4] explicitly and show that they coincide.

The lower bound $n \geq 5$ in Theorem 1.2 is sharp. In fact, for $n \leq 4$ we will show that the algebra Λ is derived unique.

The paper is organized as follows. In Section 2 we introduce the family $\Lambda(L, R)$ and record some elementary properties of the algebra $\Lambda = \Lambda(L, L)$. In Section 3 we prove Theorem 1.1 by passing to trivial extensions and using the derived equivalence classification of Brauer graph algebras together with a calculation of Cartan matrices. In Section 4 we construct Λ' , compute the surface invariants associated with Λ and Λ' , and prove Theorem 1.2. We conclude by discussing the exceptional cases $n \leq 4$.

1.1. Notations and conventions. Our notations and conventions are basically consistent with those of [3]. In particular, we assume that all modules are on the right, tilting and cotilting modules are in the classical sense [3, Definition VI.2.1], and all tilting and cotilting modules are basic. Arrows

of quivers are composed from left to right, but morphisms are from right to left. Throughout this paper, let $\mathbb{k}$ be a fixed algebraically closed field.

2. THE FAMILY $\Lambda(L, R)$

In this section, we construct a family of gentle algebras and describe their properties.

Fix an integer $n \geq 1$. Let $E = \{1, 2, \dots, n\}$, and let $L = (a_1, \dots, a_n)$ and $R = (b_1, \dots, b_n)$ be two total orderings of E . We define a bound quiver algebra $\Lambda(L, R) = \mathbb{k}Q/I$ as follows: the quiver Q is formed by vertices $1, 2, \dots, n$ and arrows $\alpha_s : a_s \rightarrow a_{s+1}$ and $\beta_s : b_s \rightarrow b_{s+1}$ ($1 \leq s \leq n-1$); the admissible ideal I is generated by all composable paths of the form $\alpha_i\beta_j$ or $\beta_j\alpha_i$. We denote the primitive idempotents of $\Lambda(L, R)$ by $e_1, \dots, e_n$. From this construction, one easily sees that $\Lambda(L, R)$ is a finite-dimensional connected gentle algebra.

Without loss of generality, we will take $L = (1, 2, \dots, n)$, the natural linear ordering of E , in what follows. We also write $\Lambda = \Lambda(L, L)$ for convenience. Therefore, we have the following explicit presentation of Λ :

$$\Lambda = \mathbb{k} \left(1 \xrightarrow[\beta_1]{\alpha_1} 2 \xrightarrow[\beta_2]{\alpha_2} \cdots \xrightarrow[\beta_{n-1}]{\alpha_{n-1}} n \right) / \langle \alpha_i\beta_{i+1}, \beta_i\alpha_{i+1} \mid 1 \leq i \leq n-2 \rangle.$$

It then follows immediately that Λ is self-dual, i.e., isomorphic to its opposite algebra. Moreover, Λ is of finite global dimension, since it contains no full-relation oriented cycle [9, Proposition 1.25].

3. TILTING-COTILTING UNIQUENESS

In this section we prove the following theorem.

Theorem 3.1. *The algebra Λ is tilting-cotilting unique for every integer $n \geq 1$.*

Here, by “tilting-cotilting unique” we mean that for any tilting module or cotilting module T over Λ , there is an algebra isomorphism $\text{End}_\Lambda(T) \cong \Lambda$. Since $\Lambda \cong \Lambda^{\text{op}}$, it suffices to prove only the case where T is tilting.

3.1. Cartan matrices. The proof involves two different ways of calculating Cartan matrices of algebras of the form $\Lambda(L, R)$. One is by direct computation based on the definition, as the following lemma illustrates.

Lemma 3.2. *For the algebra $\Lambda(L, R)$, its Cartan matrix $\mathbf{C}_{\Lambda(L, R)} = (c_{ij})_{n \times n}$ is given by $c_{ii} = 1$ for $1 \leq i \leq n$ and*

$$c_{ji} = \left\{ \begin{array}{cc} i <_L j, & 1; \\ i \geq_L j, & 0 \end{array} \right\} + \left\{ \begin{array}{cc} i <_R j, & 1; \\ i \geq_R j, & 0 \end{array} \right\} \quad \text{for } i \neq j.$$

In particular,

$$\mathbf{C}_\Lambda = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 2 & 1 & 0 & \cdots & 0 & 0 \\ 2 & 2 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 2 & 2 & 2 & \cdots & 1 & 0 \\ 2 & 2 & 2 & \cdots & 2 & 1 \end{bmatrix}.$$

Proof. By definition, $c_{ji} = \dim_{\mathbb{k}} e_i \Lambda e_j$. In our case, it equals the number of non-zero paths from i to j in Λ . If $i = j$, there is only one trivial path. If $i \neq j$, any non-zero non-trivial path in Λ must be formed entirely by arrows $\alpha_1, \dots, \alpha_{n-1}$, or be formed entirely by arrows $\beta_1, \dots, \beta_{n-1}$. The first case is equivalent to $i <_L j$, and the second case is equivalent to $i <_R j$. So our formula holds. $\square$

Corollary 3.3. *If Λ and $\Lambda(U, V)$ share the same Cartan matrix modulo 2, they are isomorphic.*

Proof. By the above lemma, for $i \neq j$ in E we have $i <_U j$ if and only if $i <_V j$. This implies that the two orderings U and V are identical. After relabelling, we clearly have $\Lambda(U, U) \cong \Lambda(L, L)$, hence $\Lambda \cong \Lambda(U, V)$. $\square$

3.2. Dimension vectors of tilting modules. The second calculation is subtler. As we will show, it explores a certain symmetry of the algebra Λ and its tilting modules.

Define an algebra endomorphism $\sigma : \Lambda \rightarrow \Lambda$ by the condition that σ fixes every primitive idempotent e_i ($1 \leq i \leq n$) and swaps every pair of arrows α_i, β_i for $1 \leq i \leq n-1$. It is easy to see that σ is an algebra automorphism. For any Λ -module M , denote by M^σ the module M equipped with σ -twisted right Λ -action. We denote the elements of M^σ by m^σ ; then $m^\sigma x = m\sigma(x)$ by definition.

Lemma 3.4. *Every indecomposable summand M of a tilting module T is σ -invariant, i.e., $M \cong M^\sigma$ as Λ -modules.*

Proof. Every tilting module is a support τ -tilting module [1, Proposition 2.2(b)]. Since Λ is gentle, its support τ -tilting graph $\mathcal{H}(\text{s}\tau\text{-tilt}\Lambda)$ is connected [7]. Therefore, there is a sequence of mutations from the regular module Λ to T in $\mathcal{H}(\text{s}\tau\text{-tilt}\Lambda)$, which may be identified by [1, Theorem 3.2, Corollary 3.9] with a sequence of mutations from $\Lambda[0]$ to $P_T^\bullet$, the minimal projective resolution of $T[0]$, in $\mathcal{H}(\text{2-silt}\Lambda)$, the 2-term siltting quiver. Clearly σ induces a triangulated automorphism of $\mathbf{D}^b(\Lambda\text{-mod})$, which is also denoted by σ , and every indecomposable summand of $\Lambda[0]$ is σ -invariant. Thus, by [6, Proposition 2.1] and induction on the length of the mutation sequence, we see that every indecomposable summand of $P_T^\bullet$ is also σ -invariant. Taking homology at degree 0, we see that every indecomposable summand M of T is σ -invariant. $\square$

Denote by $\mathbb{F}_2$ the finite field with two elements 0, 1. Denote by $\varepsilon_1, \dots, \varepsilon_n$ the standard basis of the n -dimensional vector space $\mathbb{F}_2^n$. The dimension vector of a module M is denoted by $\underline{\dim}_{\mathbb{k}}(M)$.

Lemma 3.5. *Let T be a tilting Λ -module, and let $T_1, \dots, T_n$ be all of its indecomposable summands. Then, up to permutation, we have*

$$\underline{\dim}_{\mathbb{k}}(T_i) \equiv \varepsilon_i \pmod{2}.$$

Proof. By [5, Section 3], indecomposable modules over Λ are in one-one correspondence with equivalence classes of strings and bands over Λ . Since band modules admit non-zero self-extensions [5, Page 165] but T is tilting, the T_i are not band modules. Let u_i be the string defining T_i , so $T_i \cong M(u_i)$. From $T_i \cong T_i^\sigma$ we obtain $M(u_i) \cong M(u_i)^\sigma = M(\sigma(u_i))$, so $\sigma(u_i) = u_i$ or u_i^{-1} . In the first case, u_i is a trivial string since no arrow is σ -invariant. In the second case, u_i is of even length, and its second half is determined by the first half, hence of the form $p\sigma(p)^{-1}$ for some string p . In both cases, it is true that $\underline{\dim}_{\mathbb{k}}(T_i) \equiv \varepsilon_j \pmod{2}$ for some j , because a vertex in the first half of u_i along with the corresponding vertex in the second half together contribute twice, but the middle vertex, say j , contributes only once. It remains to show that the correspondence $i \mapsto j$ is bijective.

The fact that $\underline{\dim}_{\mathbb{k}}(T_i)$ ($i = 1, \dots, n$) form a $\mathbb{Z}$ -basis of the Grothendieck group $K_0(\Lambda) \cong \mathbb{Z}^n$ [3, proof of Proposition 4.5] implies that $\underline{\dim}_{\mathbb{k}}(T_i) \pmod{2}$ ($i = 1, \dots, n$) form a $\mathbb{F}_2$ -basis of $\mathbb{F}_2^n$. Therefore the indices j occurring above are all distinct, and after reindexing we obtain $\underline{\dim}_{\mathbb{k}}(T_i) \equiv \varepsilon_i \pmod{2}$. $\square$

Corollary 3.6. *If $\Lambda' \cong \text{End}_\Lambda(T)$ for some tilting module T over Λ , then the Cartan matrix $\mathbf{C}_{\Lambda'} \equiv I_n \pmod{2}$.*

Proof. Recall a formula: if $\langle -, - \rangle$ is the Euler characteristic of Λ [3, Definition III.3.11], then $(\mathbf{C}_{\Lambda'})_{ji} = \langle \underline{\dim}_{\mathbb{k}}(T_i), \underline{\dim}_{\mathbb{k}}(T_j) \rangle$. Indeed, the indecomposable projective summands of Λ' are $\text{Hom}_\Lambda(T, T_i)$ ($i = 1, \dots, n$), so

$$\begin{aligned} (\mathbf{C}_{\Lambda'})_{ji} &= \dim_{\mathbb{k}} \text{Hom}_{\Lambda'}(\text{Hom}_\Lambda(T, T_j), \text{Hom}_\Lambda(T, T_i)) \\ & \quad ([3, \text{Definition 3.7, Lemma I.4.2}]) \\ &= \dim_{\mathbb{k}} \text{Hom}_\Lambda(T_j, T_i) \quad ([3, \text{Lemma VI.3.2}]) \\ &= \sum_{r=0}^{\infty} (-1)^r \dim_{\mathbb{k}} \text{Ext}_\Lambda^r(T_j, T_i) \quad (T \text{ is tilting}) \\ &= \langle \underline{\dim}_{\mathbb{k}}(T_j), \underline{\dim}_{\mathbb{k}}(T_i) \rangle. \quad ([3, \text{Proposition III.3.13}]) \end{aligned}$$

Since the matrix of $\langle -, - \rangle$ is $\mathbf{C}_\Lambda^{-T}$ and $\mathbf{C}_\Lambda^{-T} \equiv I_n \pmod{2}$ by Lemma 3.2, we see that $(\mathbf{C}_{\Lambda'})_{ji} \equiv \langle \varepsilon_j, \varepsilon_i \rangle \equiv \delta_{i,j} \pmod{2}$ by Lemma 3.5, and so $\mathbf{C}_{\Lambda'} \equiv I_n \pmod{2}$ as claimed. $\square$

3.3. Proof of Theorem 3.1. Now we can prove Theorem 3.1. The idea is roughly as follows: pass to their trivial extensions, which are Brauer graph

algebras; show that $\Lambda' = \text{End}_\Lambda(T)$ is also of the form $\Lambda(U, V)$; apply the above lemmas to show that $\Lambda \cong \Lambda'$.

Before the proof, we note that when $n = 1$ the algebra Λ is isomorphic to $\mathbb{k}$ and the conclusion is trivial. Since the classification theorem [10, Theorem 7.12] excludes local algebras, we assume $n \geq 2$ in the following proof.

Proof of Theorem 3.1. Let $\Lambda' = \text{End}_\Lambda(T)$, where T is a tilting Λ -module. Λ' is gentle by [2, Theorem 8.1]. Denote $B = \text{Triv}(\Lambda)$, the trivial extension of Λ ; similarly denote B' for Λ' . They are both Brauer graph algebras by [12, Theorem 1.2], so they are defined by Brauer graphs Γ and Γ' , respectively. Moreover, Λ and Λ' are obtained from them respectively by admissible cuts along the arrows closing the maximal permitted paths; see [12, Section 4] for relevant definitions and theorems. A moment's calculation shows that Γ has the following properties: it has 2 vertices both of multiplicity 1, and n edges connecting them, thus bipartite.

Since Λ and Λ' are derived equivalent, so are B and B' [11, Theorem 3.1]. By the criterion for derived equivalence between Brauer graph algebras [10, Theorem 7.12], we see that Γ' shares the same properties as Γ as above. More precisely, Γ' is bipartite, so it has no loops and thus each of its n edges joins the two distinct vertices. Therefore, Γ' defines two cyclic orderings of $E = \{1, \dots, n\}$, and they turn into two total orderings U, V of E by an admissible cut yielding Λ' , thus $\Lambda' \cong \Lambda(U, V)$.

Finally, since $\mathbf{C}_{\Lambda'} \equiv I_n \pmod{2}$ by Corollary 3.6, the result now follows from Lemma 3.3 as we have just shown that Λ' is of the form $\Lambda(U, V)$. $\square$

Remark 3.7. The first two paragraphs of the proof of Theorem 3.1 imply that the class of gentle algebras of the form $\Lambda(L, R)$ is closed under derived equivalence.

4. NON-DERIVED UNIQUENESS

For $n \geq 5$ we take $\Lambda' = \Lambda(L, R)$ where

$$R = (5, 6, \dots, n, 1, 2, 3, 4).$$

We are going to prove the following theorem.

Theorem 4.1. *For $n \geq 5$, the algebras Λ and Λ' are derived equivalent but not isomorphic. In particular, Λ is not derived unique.*

That $\Lambda \not\cong \Lambda'$ is easy to see by comparing their Cartan matrices. Alternatively, notice that the projective Λ -module at vertex n is simple, whereas Λ' has no simple projective module.

To check that they are derived equivalent, we will compare the numerical invariants introduced in [2, Theorem 7.4] on the dissected surfaces (S, M, P, Δ) and (S', M', P', Δ') associated with them, respectively. [2, Theorem 7.4] is divided into three parts. We check them one by one.

4.1. The ribbon surface. We first describe the dissected surfaces.

In general, the algebra $\Lambda(U, V)$ has precisely two maximal permitted paths: the path consisting entirely of α -arrows and the path consisting entirely of β -arrows. Consequently, its ribbon graph has two vertices, which we denote by v_1 and v_2 , and n edges $e_1, \dots, e_n$ connecting them. The cyclic ordering at v_1 is induced by U , and that at v_2 is induced by V .

In our case, L and R induce the same cyclic ordering $1 < 2 < \dots < n < 1$. Hence Λ and Λ' have the same underlying ribbon graph and therefore we shall identify S and S' . But the positions of marked points are different. More precisely, the two $\circ$ -marked points on S both lie between 1 and n , but on S' , one is between 1 and n , the other is between 4 and 5.

Lemma 4.2. *With notation as in [2, Theorem 7.4], we have*

$$g = g' = \begin{cases} \frac{n-1}{2}, & n \text{ odd}; \\ \frac{n}{2} - 1, & n \text{ even}, \end{cases} \quad b = b' = \begin{cases} 1, & n \text{ odd}; \\ 2, & n \text{ even}, \end{cases}$$

$$\#M = \#M' = 4, \quad \#P = \#P' = 0.$$

Moreover, if n is odd, the two $\circ$ -marked points lie on the unique boundary component, while if n is even, the two $\circ$ -marked points lie on distinct boundary components, both for Λ and for Λ' .

Proof. The numbers of $\circ$ -marked points and $\bullet$ -marked points both equal 2. Since there is no full-relation oriented cycle in Λ and Λ' , there is no puncture on S and S' , so $P = P' = \emptyset$.

The number of boundary components equals the number of faces [9, Definition 1.4] on the ribbon graph Γ_Λ associated with Λ . A direct counting yields the formula for b and b' . Finally, employing the formula

$$n = \#M_\circ + \#P + b + 2g - 2,$$

in [2, Proposition 3.12], one finds the genera g and g' as stated.

The positions of $\circ$ -marked points are easy to see. □

4.2. Boundary winding numbers. When n is odd, take a simple closed curve c on S which is homotopic to the unique boundary component, with orientation induced from S . When n is even, take two simple closed curves c_1, c_2 on S which are homotopic to the two boundary components, respectively, with induced orientation.

Lemma 4.3. *The boundary winding numbers for Λ and Λ' coincide. More precisely,*

- (1) *When n is odd, $w^{\Delta^*}(c) = 4 - 2n$.*
- (2) *When n is even, $w^{\Delta^*}(c_1) = w^{\Delta^*}(c_2) = 2 - n$.*

Proof. (1) When n is odd, there is only one boundary component of S , on which there are two $\bullet$ -marked points, and there are n $\bullet$ -arcs connecting one to the other. So the winding number is $4 - 2n$ by [2, Lemma 3.18].

(2) When n is even, there are two boundary components of S , on each of which there is one $\bullet$ -marked point, and there are n $\bullet$ -arcs connecting one to the other. So the winding numbers are both $2 - n$, by the same counting strategy as above.

The same calculation holds for Λ' . $\square$

4.3. A convenient family of closed curves. Having compared the topological data and boundary winding numbers of two marked surfaces, we continue with [2, Theorem 7.4(3)]. Since $g = g' \geq 2$, we are in case (b) of [2, loc. cit.].

By embedding the ribbon graph Γ_Λ into S as a deformation retract, we can view the edges $e_1, \dots, e_n$ as $\circ$ -arcs on S . We now introduce a convenient collection of simple closed curves on S . For $i \neq j$, let γ_{ij} be the simple closed curve on S based at v_1 , formed by first going along e_i and returning along e_j . It is clear that the γ_{ij} generate the fundamental group $\pi_1(S)$, and using [2, Lemma 3.18] we have

$$\begin{aligned} w^{\Delta^*}(\gamma_{ij}) &= 0, \quad \forall i, j, \\ w^{\Delta'^*}(\gamma_{ij}) &= \begin{cases} 0, & \text{if } L, R \text{ give the same order to } i, j; \\ \pm 2, & \text{if } L, R \text{ give opposite orders to } i, j. \end{cases} \end{aligned} \quad (4.1)$$

Lemma 4.4. *For every simple closed curve γ on S , the winding numbers $w^{\Delta^*}(\gamma)$ and $w^{\Delta'^*}(\gamma)$ are even.*

Proof. We prove the lemma for $w^{\Delta^*}(\gamma)$ only, because for $w^{\Delta'^*}(\gamma)$ the proof is similar. By [2, Proposition 3.4(4)], $w^{\Delta^*}(\gamma)$ factors through an element of the cohomology group $H^1(S; \mathbb{F}_2)$, which is isomorphic to the dual vector space $H_1(S; \mathbb{F}_2)^*$. So there is a linear functional $f : H_1(S; \mathbb{F}_2) \rightarrow \mathbb{F}_2$ and a commutative diagram

$$\begin{array}{ccc} \pi_1^{\text{free}}(S) & \xrightarrow{w^{\Delta^*}} & \mathbb{Z} \\ \downarrow & & \downarrow q \\ H_1(S; \mathbb{F}_2) & \xrightarrow{f} & \mathbb{F}_2, \end{array}$$

the vertical mappings being canonical.

To show $\text{Im } w^{\Delta^*} \subseteq 2\mathbb{Z}$ it suffices to show that $f = 0$ in the above diagram. First, we compute $f([\gamma_{ij}]) = q(w^{\Delta^*}(\gamma_{ij})) = q(\text{an even integer}) = 0$. Then, notice that the homology classes $[\gamma_{ij}]$ span $H_1(S; \mathbb{F}_2)$, so $f = 0$ as expected. $\square$

When $n \equiv 2 \pmod{4}$, we are in case (ii) by Lemma 4.3, and so Λ and Λ' are derived equivalent.

4.4. The quadratic form and the intersection form. Now assume that n is odd or $n \equiv 0 \pmod{4}$. We are now in case (iii), and we must compute the Arf invariants of Λ and Λ' .

In our situation, [2, Lemma 7.5] is applicable and, to begin with, let us describe the homology group $H_1(S; \mathbb{F}_2)$. It is a $\mathbb{F}_2$ -vector space of dimension $n - 1$ and we take $[\gamma_{1n}], \dots, [\gamma_{n-1,n}]$ as its basis. It is canonically isomorphic to the vector space with $\{A \subseteq E \mid \#A \text{ is even}\}$ as the underlying set and symmetric difference as addition; an isomorphism is explicitly given by $[\gamma_{in}] \mapsto \{i, n\}$, $i = 1, \dots, n - 1$. We denote the latter vector space by V and we shall use it later as an alternative description of $H_1(S; \mathbb{F}_2)$.

Next, we must calculate the quadratic form and the intersection form $(-, -)$ on $H_1(S; \mathbb{F}_2)$ mentioned in [2, Lemma 7.5]. Let us do it for Λ first. There is a unique quadratic form $q_{\Delta^*} : H_1(S; \mathbb{F}_2) \rightarrow \mathbb{F}_2$, such that

$$q_{\Delta^*}(x + y) = q_{\Delta^*}(x) + q_{\Delta^*}(y) + (x, y), \quad \forall x, y \in H_1(S; \mathbb{F}_2);$$

$$q_{\Delta^*}([\gamma_{ij}]) = \frac{1}{2} w^{\Delta^*}(\gamma_{ij}) + 1 \stackrel{(4.1)}{=} 1, \quad \forall i, j \ (i \neq j).$$

For Λ' , a similar calculation shows that the quadratic form $q_{\Delta'^*} : H_1(S; \mathbb{F}_2) \rightarrow \mathbb{F}_2$ satisfies

$$q_{\Delta'^*}(x + y) = q_{\Delta'^*}(x) + q_{\Delta'^*}(y) + (x, y), \quad \forall x, y \in H_1(S; \mathbb{F}_2);$$

$$\begin{aligned} q_{\Delta'^*}([\gamma_{ij}]) &= \frac{1}{2} w^{\Delta'^*}(\gamma_{ij}) + 1 \\ &= \begin{cases} 1, & \text{if } L, R \text{ give the same order to } i, j; \\ 0, & \text{if } L, R \text{ give opposite orders to } i, j, \end{cases} \quad \forall i, j \ (i \neq j). \end{aligned}$$

Lemma 4.5. (1) If n is odd, $(-, -)$ is non-degenerate;

(2) If $n \equiv 0 \pmod{4}$, $(-, -)$ is degenerate, and its radical is 1-dimensional, spanned by $[\gamma_{1n}] + \dots + [\gamma_{n-1,n}]$, or alternatively, spanned by $E \in V$ via the canonical isomorphism.

Proof. For later use, let us write

$$J_n = \begin{bmatrix} 0 & 1 & \cdots & 1 & 1 \\ 1 & 0 & \cdots & 1 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & \cdots & 0 & 1 \\ 1 & 1 & \cdots & 1 & 0 \end{bmatrix} \in \mathbb{F}_2^{n \times n}.$$

Let us calculate the matrix of $(-, -)$ under the basis $[\gamma_{1n}], \dots, [\gamma_{n-1,n}]$. For Λ and $1 \leq i \neq j < n$ we have

$$\begin{aligned} ([\gamma_{in}], [\gamma_{jn}]) &= q_{\Delta^*}([\gamma_{in}] + [\gamma_{jn}]) + q_{\Delta^*}([\gamma_{in}]) + q_{\Delta^*}([\gamma_{jn}]) \\ &= q_{\Delta^*}([\gamma_{ij}]) + q_{\Delta^*}([\gamma_{in}]) + q_{\Delta^*}([\gamma_{jn}]) \\ &= 1 + 1 + 1 = 1. \end{aligned}$$

So the matrix of $(-, -)$ is J_{n-1} .

By linear algebra, we see that the matrix J_{n-1} is non-degenerate if n is odd, and is degenerate if n is even. In the latter case, its radical is the 1-dimensional subspace spanned by the vector $(1, 1, \dots, 1)^T$. Thus (1) and (2) follow. $\square$

Remark 4.6. The intersection form $(-, -)$ is an intrinsic topological pairing on $H_1(S; \mathbb{F}_2)$ and is independent of the particular dissection. However, the two quadratic forms q_{Δ^*} and $q_{\Delta'^*}$ are not equal, as is shown by explicit formulae above.

4.5. The Arf invariant. According to [2, Lemma 7.5], the forms q_{Δ^*} and $q_{\Delta'^*}$ descend to quadratic forms $\bar{q}_{\Delta^*}$ and $\bar{q}_{\Delta'^*}$, respectively, on the closed surface $\bar{S}$ obtained from S by capping the boundary components by discs. The inclusion map $S \rightarrow \bar{S}$ induces a long exact sequence of homology groups:

$$0 \rightarrow H_2(\bar{S}; \mathbb{F}_2) \xrightarrow{\cong \mathbb{F}_2} H_2(\bar{S}, S; \mathbb{F}_2) \xrightarrow{\cong \begin{cases} \mathbb{F}_2, & n \text{ odd}, \\ \mathbb{F}_2^2, & n \text{ even} \end{cases}} H_1(S; \mathbb{F}_2) \xrightarrow{\cong \mathbb{F}_2^{n-1}} H_1(\bar{S}; \mathbb{F}_2) \xrightarrow{\cong \mathbb{F}_2^{2g}} H_1(\bar{S}, S; \mathbb{F}_2) \xrightarrow{\cong 0} 0.$$

We see that: if n is odd then $H_1(S; \mathbb{F}_2) \cong H_1(\bar{S}; \mathbb{F}_2)$; if n is even then the induced map $H_1(S; \mathbb{F}_2) \rightarrow H_1(\bar{S}; \mathbb{F}_2)$ is surjective and realizes $H_1(\bar{S}; \mathbb{F}_2)$ as the quotient space of $H_1(S; \mathbb{F}_2)$ by the radical of $(-, -)$. So $(-, -)$ descends to a form $\overline{(-, -)}$ on $H_1(\bar{S}; \mathbb{F}_2)$.

By [2, Lemma 7.5], the Arf invariants of Λ and Λ' are calculated by first specifying a symplectic geometric basis $\bar{u}_1, \bar{v}_1, \dots, \bar{u}_g, \bar{v}_g$ of $H_1(\bar{S}; \mathbb{F}_2)$, then calculating the sums

$$\text{Arf}(\Lambda) = \sum_{i=1}^g \bar{q}_{\Delta^*}(\bar{u}_i) \bar{q}_{\Delta^*}(\bar{v}_i), \quad \text{Arf}(\Lambda') = \sum_{i=1}^g \bar{q}'_{\Delta^*}(\bar{u}_i) \bar{q}'_{\Delta^*}(\bar{v}_i).$$

In order to do this, we take vectors

$$\begin{aligned} u_i &= (0, \dots, 0, \overset{2i-1}{1}, \overset{2i}{1}, 0, \dots, \overset{n-1}{0}), \\ v_i &= (1, \dots, 1, \overset{2i-1}{1}, \overset{2i}{0}, 0, \dots, \overset{n-1}{0}), \end{aligned} \quad (i = 1, \dots, g)$$

with respect to the basis $[\gamma_{1n}], \dots, [\gamma_{n-1,n}]$, and denote by $\bar{u}_1, \bar{v}_1, \dots, \bar{u}_g, \bar{v}_g$ their images in $H_1(\bar{S}; \mathbb{F}_2)$. One can verify that they indeed form a symplectic basis under $\overline{(-, -)}$. Passing to the $\mathbb{F}_2$ -vector space V introduced in Subsection 4.4, u_i becomes $\{2i-1, 2i\}$ and v_i becomes $\{1, 2, \dots, 2i-1, n\}$. Let μ_i be the simple closed curve on S formed by first going along e_{2i-1} then returning along the inverse of e_{2i} , or simply write $\mu_i = e_{2i-1} e_{2i}^{-1}$. Similarly, let ν_i be the simple closed curve on S expressed by $e_1 e_2^{-1} \cdots e_{2i-1} e_n^{-1}$. Then μ_i, ν_i are simple closed curves on S representing the homology classes u_i, v_i , respectively. So $\bar{u}_1, \bar{v}_1, \dots, \bar{u}_g, \bar{v}_g$ is in fact a symplectic geometric basis.

Lemma 4.7. *The Arf invariants of Λ and Λ' are equal.*

Proof. Since $\bar{q}_{\Delta^*}$ and $\bar{q}_{\Delta'^*}$ are descended from q_{Δ^*} and $q_{\Delta'^*}$, respectively, we have

$$\text{Arf}(\Lambda) = \sum_{i=1}^g q_{\Delta^*}(u_i) q_{\Delta^*}(v_i), \quad \text{Arf}(\Lambda') = \sum_{i=1}^g q_{\Delta'^*}(u_i) q_{\Delta'^*}(v_i).$$

Using formulae in Subsection 4.4, one finds that

$$\begin{aligned} q_{\Delta^*}(u_i) &= 1, & q_{\Delta^*}(v_i) &= i, & \forall i \geq 1, \\ q_{\Delta'^*}(u_i) &= 1, & & & \forall i \geq 1, \\ q_{\Delta'^*}(v_i) &= i, & & & \forall i \geq 3, \end{aligned}$$

with exception

$$q_{\Delta'^*}(v_1) = 0, \quad q_{\Delta'^*}(v_2) = 1.$$

But both Arf invariants turn out to be $\frac{g(g+1)}{2} \pmod{2}$. $\square$

Now, Theorem 4.1 is immediate.

Proof of Theorem 4.1. For all $n \geq 5$, [2, Theorem 7.4(1)(2)] holds by Lemma 4.2, 4.3. When $n \equiv 2 \pmod{4}$, [2, Theorem 7.4(3)(b)(ii)] holds by the discussion in Subsection 4.3. When n is odd or $n \equiv 0 \pmod{4}$, [2, Theorem 7.4(3)(b)(iii)] follows from Lemma 4.7. $\square$

Combining Theorem 4.1 and 3.1, we obtain the answer to our problem:

Corollary 4.8. *There exist two gentle algebras which are derived equivalent but not tilting-cotilting equivalent.*

4.6. Remark on exceptional cases. We note that for $n \leq 4$, the gentle algebra Λ is derived unique. The cases $n = 1, 2$ are obvious. The case $n = 3$ is contained in [4, Theorem A]. For the case $n = 4$, we make it a proposition.

Proposition 4.9. *If $n = 4$, the gentle algebra Λ is derived unique.*

Proof. Assume that Λ' is derived equivalent to Λ . By Remark 3.7, we may take $\Lambda' = \Lambda(L, R)$, and let $L = (1, 2, 3, 4)$. If we view L, R as permutations on $E = \{1, 2, 3, 4\}$ as they are written, then the faces of the ribbon graph $\Gamma_{\Lambda'}$ associated with Λ' are in bijection with the cycles of LR , and the perimeter (= number of boundary segments) of a face equals the length of the corresponding cycle. Since Γ_{Λ} has precisely two faces with perimeter 2, so must do $\Gamma_{\Lambda'}$, and LR must decompose as a product of two disjoint transpositions. Among all 4-cycles on E the only one satisfying this condition is (1234). Therefore there are only 4 possibilities for R : $R = (1, 2, 3, 4)$, $(2, 3, 4, 1)$, $(3, 4, 1, 2)$ or $(4, 1, 2, 3)$.

If $R = (2, 3, 4, 1)$ or $(4, 1, 2, 3)$, then Λ' is of infinite global dimension since it contains a full-relation oriented cycle. But Λ is of finite global dimension, so these two possibilities are thus eliminated.

If $R = (3, 4, 1, 2)$, we compare their numerical invariants as before. Both dissected surfaces are of genus 1, so we are in case (a) of [2, Theorem 7.4(3)]. A calculation similar to that in Lemma 4.3 shows that all boundary winding numbers are -2 . A calculation similar to the one leading to Equation (4.1) shows that for Λ , all winding numbers of curves γ_{ij} ($1 \leq i, j \leq 4$) are 0, so we obtain $\gcd(0, 0, -2 + 2, -2 + 2) = 0$. For Λ' , notice that γ_{13}, γ_{14} are simple closed curves intersecting transversely at only one point, so they form a symplectic geometric basis of $H_1(\bar{S}; \mathbb{F}_2)$ and we obtain $\gcd(\pm 2, \pm 2, -2 +$

$2, -2 + 2) = 2$. The numerical invariants disagree, so Λ and Λ' are not derived equivalent.

Therefore, $R = L$ and $\Lambda' \cong \Lambda$. Thus Λ is derived unique. $\square$

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